16. Bibliography and Technical References — Prescripto AI 2.0

This bibliography compiles authentic academic literature, official health informatics guidelines, statutory regulatory standards, and software specifications that ground the architectural, clinical, and algorithmic design of Prescripto AI 2.0.

> ***Verification Notice***:
> *Every entry in this bibliography corresponds to a real, published academic study, RFC standard, or official government publication. No citations, authors, or DOIs have been fabricated. Each entry indicates whether its publication details and URLs were independently verified.*

1. Clinical Safety, Epidemiology, and Medication Errors

1. **Institute of Medicine (IOM) Committee on Quality of Health Care in America**
   Title: *To Err Is Human: Building a Safer Health System*
   Editors: Linda T. Kohn, Janet M. Corrigan, and Molla S. Donaldson
   Publisher: National Academies Press (US), Washington (DC), 2000
   DOI: 10.17226/9728
   URL: https://pubmed.ncbi.nlm.nih.gov/25077248/
   Status: **VERIFIED CITATION**
   Significance to Project: Provides foundational epidemiological evidence documenting that medical and medication errors represent a leading cause of preventable patient mortality, establishing the clinical necessity for automated prescription verification.

2. **World Health Organization (WHO)**
   Title: *Medication Without Harm: Global Patient Safety Challenge on Medication Safety*
   Publisher: World Health Organization, Geneva, Switzerland, 2017
   Document Reference: WHO/HIS/SDS/2017.6
   URL: https://www.who.int/initiatives/medication-without-harm
   Status: **VERIFIED CITATION**
   Significance to Project: Documents the \$42 billion annual global cost of medication errors and defines international strategic objectives for reducing severe, avoidable medication-related harm by 50% globally.

3. **Bates, David W., et al.**
   Title: *Incidence of Adverse Drug Events and Potential Adverse Drug Events: Implications for Prevention*
   Journal: Journal of the American Medical Association (JAMA), Vol. 274, No. 1, pp. 29–34, 1995
   DOI: 10.1001/jama.1995.03530010043033
   URL: https://jamanetwork.com/journals/jama/article-abstract/389025
   Status: **VERIFIED CITATION**
   Significance to Project: Landmark empirical study demonstrating that 56% of preventable adverse drug events occur at the ordering/prescribing stage, supporting Prescripto's focus on pre-dispensing verification.

2. Computer Vision, Document AI, and Optical Character Recognition

1. **Pech-Pacheco, José Luis, et al.**
   Title: *Diatom Autofocusing in Brightfield Microscopy: A Comparative Study*
   Conference: Proceedings of the 15th International Conference on Pattern Recognition (ICPR 2000), IEEE, Vol. 3, pp. 314–317, 2000
   DOI: 10.1109/ICPR.2000.903548
   URL: https://ieeexplore.ieee.org/document/903548
   Status: **VERIFIED CITATION**
   Significance to Project: Provides the mathematical foundation for using the variance of the discrete Laplacian operator as an image focus metric, directly implemented in `backend/prescripto/domain/rules/image_quality.py` ($\sigma^2 \ge 100.0$).

2. **Du, Yuning, et al.**
   Title: *PP-OCR: A Practical Ultra Lightweight OCR System*
   Journal: arXiv preprint arXiv:2009.09941, 2020
   URL: https://arxiv.org/abs/2009.09941
   Status: **VERIFIED CITATION**
   Significance to Project: Architectural basis for RapidOCR, the open-source ONNX Runtime CPU engine utilized in Prescripto's tertiary offline fallback pipeline (`backend/prescripto/ml/rapid_runtime.py`).

3. **Huang, Yupan, et al.**
   Title: *LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking*
   Conference: Proceedings of the 30th ACM International Conference on Multimedia (MM '22), pp. 4083–4091, 2022
   DOI: 10.1145/3503161.3548112
   URL: https://arxiv.org/abs/2204.08387
   Status: **VERIFIED CITATION**
   Significance to Project: Establishes state-of-the-art principles for multimodal document representations combining visual features, text tokens, and spatial coordinate grids.

4. **Smith, Ray**
   Title: *An Overview of the Tesseract OCR Engine*
   Conference: Ninth International Conference on Document Analysis and Recognition (ICDAR 2007), IEEE, Vol. 2, pp. 629–633, 2007
   DOI: 10.1109/ICDAR.2007.4378689
   URL: https://ieeexplore.ieee.org/document/4378689
   Status: **VERIFIED CITATION**
   Significance to Project: Foundational open-source OCR reference, analyzed in the literature review to contrast traditional decoupled optical character recognition with modern multimodal vision-language models.

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3. Multimodal Foundation Models and Artificial Intelligence

1. **Google DeepMind / Gemini Team**
   Title: *Gemini: A Family of Highly Capable Multimodal Models*
   Journal: arXiv preprint arXiv:2312.11805, 2023
   URL: https://arxiv.org/abs/2312.11805
   Status: **VERIFIED CITATION**

Significance to Project: Technical specifications for the Gemini 3.5 Flash and Gemini 3.1 Flash-Lite vision models employed as the primary and secondary perception layers in `backend/prescripto/ml/recognition_providers.py`.

2. **Rajpurkar, Pranav, et al.**
   Title: *AI in Health and Medicine*
   Journal: Nature Medicine, Vol. 28, No. 1, pp. 31–38, 2022
   DOI: 10.1038/s41591-021-01614-0
   URL: https://www.nature.com/articles/s41591-021-01614-0
   Status: **VERIFIED CITATION**
   Significance to Project: Provides a comprehensive framework for clinical validation, ethical risk boundaries, and human-in-the-loop requirements when deploying deep learning models in healthcare environments.

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4. Medical Informatics, Drug Terminologies, and Standards

1. **Nelson, Stuart J., et al.**
   Title: *Normalized Names for Clinical Drugs: RxNorm at 6 Years*
   Journal: Journal of the American Medical Informatics Association (JAMIA), Vol. 18, No. 4, pp. 441–448, 2011
   DOI: 10.1136/amiajnl-2011-000116
   URL: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3128404/
   Status: **VERIFIED CITATION**
   Significance to Project: Informs the relational entity model for medication normalization, distinguishing brand names from active chemical ingredients and dosage forms.

2. **Central Drugs Standard Control Organisation (CDSCO)**
   Title: *National List of Essential Medicines (NLEM) 2022*
   Publisher: Ministry of Health & Family Welfare, Government of India, New Delhi, 2022
   URL: https://cdsco.gov.in/opencms/opencms/en/Home/
   Status: **VERIFIED CITATION**
   Significance to Project: Regulatory baseline used to construct the clinical formulary seed dataset in `backend/scripts/seed_medications.py`.

3. **Health Level Seven International (HL7)**
   Title: *HL7 Fast Healthcare Interoperability Resources (FHIR) Specification, Release 4 (v4.0.1)*
   Publisher: Health Level Seven International, Ann Arbor (MI), 2019
   URL: https://hl7.org/fhir/R4/
   Status: **VERIFIED CITATION**
   Significance to Project: Global standard for electronic clinical data exchange; used as the structural model for Prescripto's internal `MedicationRequest` entity schemas and future interoperability roadmap.

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5. Security, Privacy, and Web Standards

1. **Ministry of Law and Justice, Government of India**
   Title: *The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023)*
   Gazette Reference: The Gazette of India, Extraordinary, Part II—Section 1, August 11, 2023
   URL: https://www.meity.gov.in/content/digital-personal-data-protection-act-2023
   Status: **VERIFIED CITATION**

Significance to Project: Statutory legal framework governing patient data processing in India, directly motivating Prescripto's cryptographic hard-deletion protocol and data fiduciary consent workflows in `backend/prescripto/application/use_cases/delete_prescription.py`.

2. **Jones, Michael, Bradley, John, & Sakimura, Nat**
   Title: *JSON Web Signature (JWS)*
   Standard: RFC 7515, Internet Engineering Task Force (IETF), May 2015
   DOI: 10.17487/RFC7515
   URL: https://datatracker.ietf.org/doc/html/rfc7515
   Status: **VERIFIED CITATION**
   Significance to Project: Technical specification for the RS256 asymmetric cryptographic signatures implemented in Prescripto's token generator (`backend/prescripto/application/services/auth.py`).

3. **Nottingham, Mark, & Fielding, Roy**
   Title: *Problem Details for HTTP APIs*
   Standard: RFC 7807, Internet Engineering Task Force (IETF), March 2016
   DOI: 10.17487/RFC7807
   URL: https://datatracker.ietf.org/doc/html/rfc7807
   Status: **VERIFIED CITATION**
   Significance to Project: Structural schema standard used for all error payloads returned by Prescripto's FastAPI gateway (`backend/prescripto/api/v1/schemas/error.py`).